

Coupling Algebras: Gluing, the Square-Zero Cone and the Ring of Contingency Tables

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Abstract

Couplings of finite distributions carry two algebraic structures, and this note determines both. The first is the *gluing algebra* A_f : matrices whose row and column sums are multiples of a distribution f , multiplied by the gluing lemma. We show that $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$, with the product coupling $f \otimes f$ and its complement $\text{diag}(f) - f \otimes f$ as the two central idempotents. So the circulations form a direct factor, the mass functional is the projection onto the other factor, and the algebra sees nothing of positivity. Whiskering by a kernel is not multiplicative. Its defect makes it an A_∞ -functor into the square-zero cone of the circulations, and we show that this functor is the unique one with its first component, for every mass-preserving linear map, not only for whiskering. Up to homotopy it is the mass functional: the cone is quasi-isomorphic to $\mathbb{R}$, the gluing algebra is separable, and so every homotopy invariant of a functor that kills the circulations sees only the total mass. We also show that couplings with kernels as vertical arrows do not form a double category: squares compose horizontally for bijections, but not for some constant kernels or for some non-injective functions. The second structure is the *ring of contingency tables*: the Ehrhart ring of a coupling polytope with rational margins. It is generated in degree one, its Hilbert function counts integer tables with scaled margins, and its Proj is the toric variety of the polytope. Its degree-one basis is the sample space of Fisher's exact test, and the basic moves that connect it are integer circulations. The circulations thus appear three times: as the ideal of the gluing algebra, as the tangent directions of the coupling polytope, and as the Markov moves of exact conditional inference.

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*Every statement is checked by `check_coupling_algebra.py` (10 checks, exact arithmetic except where marked).

1 Introduction

The theory paper [3] linearizes the coupling 2-cells of the Kleisli category of the finite distribution monad. For a distribution f on a finite set, it forms the algebra A_f of matrices whose row and column sums are multiples of f , with the gluing product, and the ideal I_f of circulations. Whiskering by a kernel is then a linear map that is not multiplicative, and its defect is the second Boolean cumulant. Into the square-zero cone of I_f , whiskering is an A_∞ -functor with exactly two components [3, Thm. 5.2], also across objects [3, Thm. 10.2]. Two questions were left: why this cone, and whether a larger cone carries more [3, Question 9.4]. A third question, from the attribution paper [2, Open problem 5], asks for a strict structure on the witness layer.

This note answers all three, and identifies a second algebra that does carry the geometry. The answers are not the ones a reader might hope for. The cone is homotopy-initial among targets that kill the circulations, but only because it is equivalent to $\mathbb{R}$: the gluing algebra splits off its circulations as a direct factor. The functor into it is forced, and up to homotopy it remembers only the mass. And couplings with kernels do not form a double category. What does carry the coupling geometry is a different algebra, graded and commutative: the semigroup ring of lattice points of the coupling polytope.

Main results.

- (A) *Structure* (theorem 2.2 and proposition 2.3). $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$. The product coupling $P_f = f \otimes f$ and $e_f = \text{diag}(f) - P_f$ are complementary central idempotents, $I_f = e_f A_f$, and the mass is the projection onto $\mathbb{R}P_f$. Across objects the linear category of couplings splits as the indiscrete category on $\mathbb{R}$ times the category of circulations.
- (B) *Forced functor* (theorem 4.1). For every linear map F_1 that preserves mass, there is exactly one A_∞ -functor into the square-zero cone with first component F_1 : its second component is the multiplicativity defect, and all higher components vanish.
- (C) *Homotopy content* (theorem 4.2 and corollary 4.3). The cone is quasi-isomorphic to $\mathbb{R}$, and the functor composed with this quasi-isomorphism is the mass, a strict algebra map. Since A_f is separable, every A_∞ -functor out of it is homotopic to a strict one, and if it makes e_f a boundary, its homotopy class factors through the mass. So no homotopy invariant of such a functor sees contextuality.
- (D) *No double category* (proposition 5.1). With couplings as horizontal arrows, kernels as vertical arrows and squares given by pushforward, squares compose horizontally when the middle kernel is a bijection, but not for some constant kernels or some non-injective functions. The defect is a circulation.
- (E) *The ring of contingency tables* (theorem 6.1). For margins with common denominator N , the polytope $N \cdot \text{Cpl}(f, g)$ is a lattice polytope with the integer decomposition property. Its Ehrhart ring is generated in degree one, its Hilbert function counts integer tables, and its Proj is the toric variety of the coupling polytope. Its degree-one basis is the sample space of Fisher's exact test, connected by basic moves that are integer circulations.

What is not claimed. Earlier drafts of this programme proposed three stronger statements, and each is false. The square-zero cone is homotopy-initial among targets killing I_f , but it is equivalent to $\mathbb{R}$, so initiality says nothing beyond the mass. Its minimal model, in the sense of A_∞ -algebras [11], is $\mathbb{R}$ itself. And the witness layer with kernels as vertical maps is not a double category, strict or lax. The gluing algebra is also not a coordinate ring, since it is not commutative, and $A_f/I_f \cong \mathbb{R}$ is not the cohomology ring of the toric manifold, whose dimension is the number of vertices of the polytope.

Relation to the other papers. This note uses the definitions of [3, §§2, 5, 10] and of the coupling 2-cells of [2, §7]. It does not use the obstruction tower [6], the empirical material of [5], or the toric computations of [4], except to name the variety in [theorem 6.1](#).

2 The gluing algebra

Let f be a full-support distribution on $Y = \{1, \dots, n\}$, and $D = \text{diag}(f)$.

Definition 2.1. $A_f = \{a \in \mathbb{R}^{n \times n} : a\mathbf{1} = sf, \mathbf{1}^T a = sf^T \text{ for some } s = s(a) \in \mathbb{R}\}$, with product $a \cdot b = aD^{-1}b$ and unit D . The *mass* is $s(a)$, and $I_f = \ker s$ is the space of *circulations*.

Couplings of f with itself are the elements with $a \geq 0$ and $s = 1$, and their product is the gluing of [3, Thm. 2.3]. Write $P_f = ff^T$, the product coupling, and $e_f = D - P_f$.

Theorem 2.2 (Structure of the gluing algebra). *(i) P_f and e_f are idempotents with $P_f + e_f = D$ and $P_f \cdot e_f = 0$, and both are central.*

(ii) $P_f \cdot a = a \cdot P_f = s(a)P_f$ and $e_f \cdot a = a - s(a)P_f$. Hence $A_f = \mathbb{R}P_f \oplus I_f$ as algebras, $I_f = e_f A_f$ is a unital algebra with unit e_f , and s is the projection onto the first factor.

(iii) With $v = (\sqrt{f_y})_y$, the map $\varphi(a) = D^{-1/2}aD^{-1/2}$ is an isomorphism of A_f onto $\{M : Mv = \lambda v, v^T M = \lambda v^T\} = \mathbb{R}vv^T \oplus \text{End}(v^\perp)$. So $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$, a semisimple algebra, and $\dim I_f = (n-1)^2$.

Proof. (ii) $P_f D^{-1}a = f(\mathbf{1}^T a) = s(a)ff^T$, and symmetrically. Then $e_f \cdot a = a - s(a)P_f$, which has mass 0. (i) follows from (ii) with $s(P_f) = 1$ and $s(e_f) = 0$, using $f^T D^{-1}f = \sum_y f_y = 1$. (iii) $\varphi(aD^{-1}b) = \varphi(a)\varphi(b)$ and $\varphi(D) = 1$. The conditions $a\mathbf{1} = sf$ and $\mathbf{1}^T a = sf^T$ become $\varphi(a)v = sv$ and $v^T \varphi(a) = sv^T$, since $D^{1/2}\mathbf{1} = v$ and $D^{-1/2}f = v$. If $Mv = \lambda v$ and $v^T M = \lambda v^T$, then M maps $v^\perp$ into itself, so M is λvv^T plus an endomorphism of $v^\perp$; and $\varphi(P_f) = vv^T$. $\square$

So the algebra is as simple as it could be, and none of the coupling geometry is visible in it: the couplings are the nonnegative elements of mass one, and nonnegativity is not an algebraic condition here. The product coupling, the Birch point of the independence model and the Kähler class of [3, §6], appears as the central idempotent that projects onto the masses.

Several objects. Let $\mathcal{L}$ be the linear category of [3, Def. 10.1]: objects are full-support distributions on finite sets, $\mathcal{L}(f, g)$ consists of the matrices with row sums sf and column sums sg , and composition is $a \cdot b = a \text{diag}(g)^{-1}b$.

Proposition 2.3 (Splitting). *Every $a \in \mathcal{L}(f, g)$ is uniquely $a = s(a)fg^T + x$ with x a circulation, and*

$$(sf g^T + x) \cdot (tg k^T + y) = stfk^T + x \cdot y.$$

So each hom-space of $\mathcal{L}$ is the direct sum of $\mathbb{R}$ and the circulations, compatibly with composition: $\mathcal{L}$ is the componentwise sum of the linearized indiscrete category, with every hom-space $\mathbb{R}$, and the category of circulations, whose identity at f is e_f .

Proof. $fg^\top \text{diag}(g)^{-1}y = f(\mathbf{1}^\top y) = 0$ for a circulation y , and $x \text{diag}(g)^{-1}gk^\top = (x\mathbf{1})k^\top = 0$; the remaining term is $stf(\mathbf{1}^\top \text{diag}(g)^{-1}g)k^\top = stfk^\top$. $\square$

3 How kernels act

A kernel $h: Y \rightarrow D(Z)$, written as a stochastic matrix, acts on matrices in two ways [3, Def. 10.1]. The *product action* $\pi_h(a) = h^\top a h$ runs h independently on both coordinates. The *whiskering* w_h copies h on equal pairs and runs it independently on unequal pairs, which is the postcomposition of the witness layer [2, Thm. 7.6]. Both preserve mass and push margins forward.

Proposition 3.1 (The actions in split coordinates). *Write h_y for the row of h at y .*

- (i) $\pi_h(fg^\top) = (h^\top f)(h^\top g)^\top$, and π_h maps circulations to circulations. So π_h respects the splitting.
- (ii) For every a , $w_h(a) = \pi_h(a) + \sum_y a_{yy}(\text{diag}(h_y) - h_y h_y^\top)$, and each term of the sum is a circulation. In particular $w_h(P_f) = P_{h^\top f} + c_h(f)$ with the copy correction $c_h(f) = \sum_y f_y^2(\text{diag}(h_y) - h_y h_y^\top)$.
- (iii) The multiplicativity defect of the whiskering on the product coupling is $b_2(P_f, P_f) = w_h(P_f \cdot P_f) - w_h(P_f) \cdot w_h(P_f) = c_h(f) - c_h(f) \cdot c_h(f)$.

Proof. (i) is direct, and π_h of a matrix with zero row and column sums has zero row and column sums because h is stochastic. (ii) The two actions agree off the diagonal of a ; on an equal pair (y, y) copying replaces $a_{yy}h_y h_y^\top$ by $a_{yy} \text{diag}(h_y)$, and $\text{diag}(h_y) - h_y h_y^\top$ has zero row and column sums. The diagonal of P_f is $(f_y^2)_y$. (iii) $P_f \cdot P_f = P_f$, and $P_{h^\top f}$ annihilates circulations by theorem 2.2(ii). $\square$

So the whiskering mixes the two factors of theorem 2.2, and the copy correction, a weighted sum of the covariance matrices of the rows of h , measures by how much. The product action respects the factors but does not preserve the unit, since $\pi_h(D) = h^\top D h \neq \text{diag}(h^\top f)$ unless h is deterministic. This is the trade-off of [3, Thm. 10.2] seen in coordinates.

4 The square-zero cone

Let f' be a distribution and $C = A_{f'} \oplus \varepsilon I_{f'}$, with $A_{f'}$ in degree 0 and $\varepsilon I_{f'}$ in degree -1 . The product is $a \cdot \varepsilon x = \varepsilon(a \cdot x)$, $\varepsilon x \cdot a = \varepsilon(x \cdot a)$ and $\varepsilon x \cdot \varepsilon y = 0$, which is associative and unital because $I_{f'}$ is a two-sided ideal, and the differential is $m_1(\varepsilon x) = x$, which satisfies the Leibniz rule. So C is a differential graded algebra, whose underlying complex is the mapping cone of $I_{f'} \hookrightarrow A_{f'}$. We use the sign conventions of [11].

Theorem 4.1 (The functor is forced). *Let $F_1: A_f \rightarrow A_{f'}$ be linear with $s(F_1 a) = s(a)$. Then there is exactly one A_∞ -functor $F: A_f \rightarrow C$ with first component F_1 . Its components are $F_2 = \varepsilon b_2$, with $b_2(a, b) = F_1(a \cdot b) - F_1(a) \cdot F_1(b)$, and $F_n = 0$ for $n \geq 3$.*

Proof. The component F_n has degree $1 - n$. For $n \geq 3$ the cone has nothing in that degree, so $F_n = 0$. The arity-two equation reads $m_1 F_2(a, b) = b_2(a, b)$. Since F_1 preserves mass, $b_2(a, b)$ has mass $s(a)s(b) - s(a)s(b) = 0$, so it lies in $I_{f'}$. Since F_2 has degree -1 it lies in $\varepsilon I_{f'}$, where

m_1 is injective, and so $F_2 = \varepsilon b_2$ is the only solution. The arity-three equation is $F_1 a \cdot b_2(b, c) - b_2(ab, c) + b_2(a, bc) - b_2(a, b) \cdot F_1 c = 0$. Expanding b_2 , the terms $F_1(abc)$ and $F_1 a F_1 b F_1 c$ cancel, using only linearity of F_1 and associativity. In arity four the only surviving term is a product $F_2 \cdot F_2 \in \varepsilon^2 I_{f'} = 0$, and in higher arity every term contains some F_k with $k \geq 3$. $\square$

Whiskering and the product action are two instances, and so are their convex combinations. The cone is therefore not specific to whiskering: it is the receptacle of every mass-preserving linear map, which it turns into an A_∞ -functor by recording the multiplicativity defect.

Theorem 4.2 (Homotopy content). *(i) $H^0(C) = A_{f'}/I_{f'} \cong \mathbb{R}$ and $H^{-1}(C) = 0$, so the map $q(a + \varepsilon x) = s(a)$ is a quasi-isomorphism $C \rightarrow \mathbb{R}$ of differential graded algebras. In particular the minimal model of C is $\mathbb{R}$.*

(ii) For the functor F of [theorem 4.1](#), $q \circ F$ is the mass $s: A_f \rightarrow \mathbb{R}$, a strict algebra map.

(iii) Let $G: A_f \rightarrow E$ be any A_∞ -functor into an A_∞ -algebra E such that $G_1(e_f)$ is a boundary. Then $G_1(x)$ is a boundary for every circulation x , and $H(G_1)$ factors through the mass.

Proof. (i) m_1 maps $\varepsilon I_{f'}$ isomorphically onto $I_{f'}$, and $A_{f'}/I_{f'} \cong \mathbb{R}$ by [theorem 2.2](#). The map q is multiplicative because s is and because q vanishes on $\varepsilon I_{f'}$. A dg algebra with cohomology $\mathbb{R}$ in degree 0 has minimal model $\mathbb{R}$. (ii) $q(F_1 a) = s(F_1 a) = s(a)$ and $q(F_2) = 0$. (iii) An A_∞ -functor induces an algebra map on homology. For a circulation x , $x = e_f \cdot x$ by [theorem 2.2\(ii\)](#), so $H(G_1)(x) = H(G_1)(e_f) H(G_1)(x) = 0$. $\square$

Corollary 4.3. *Let $G: A_f \rightarrow E$ be an A_∞ -functor such that $G_1(e_f)$ is a boundary. Then G is homotopic, after passing to a minimal model of E , to a strict algebra map that factors through the mass s . So every homotopy invariant of G is a function of masses alone.*

Proof. By [theorem 2.2](#), A_f is a product of matrix algebras over $\mathbb{R}$, hence separable, so its Hochschild cohomology with coefficients in any bimodule vanishes in positive degrees. The obstructions to removing the higher components of an A_∞ -functor out of A_f , after passing to a minimal model of the target, lie in these groups, so G is homotopic to a strict map. That map is $H(G_1)$, which factors through s by [theorem 4.2\(iii\)](#). $\square$

In particular no such invariant detects contextuality, which is an obstruction to positivity: signed measures always glue [[5](#), Prop. 35.1], and the parity horns of [[3](#), Thm. 4.3] are compatible families with contextual fraction $1 - 2^{2-n}$ that a linear invariant cannot distinguish from noncontextual ones.

So the coupling information of the A_∞ -functor lives at chain level, in $F_2 = \varepsilon b_2$, and nowhere in its homotopy type. This settles [[3](#), Question 9.4(i)] in the negative for homotopy-invariant information. Whether the free cone carries chain-level higher components given by the higher Boolean cumulants remains open ([section 7](#)).

5 Couplings with kernels do not form a double category

Take full-support probability spaces (Y, p) as objects, couplings $\alpha \in \text{Cpl}(p, q)$ as horizontal arrows composed by gluing, and kernels k with $k^\top p = p'$ as vertical arrows. A square with horizontal arrows $\alpha \in \text{Cpl}(p, q)$ and $\alpha' \in \text{Cpl}(p', q')$ and vertical kernels k, l exists when $\alpha' = k^\top \alpha l$; squares are a property, so the structure would be a thin double category. Squares compose vertically, because π is functorial in the kernel. For them to compose horizontally, gluing must commute with pushforward along the middle kernel.

Proposition 5.1 (Interchange). *Let $\alpha \in \text{Cpl}(p, q)$, $\beta \in \text{Cpl}(q, r)$, and kernels k, l, m with l acting on the middle space. The identity*

$$k^\top(\alpha \cdot \beta)m = (k^\top\alpha l) \cdot (l^\top\beta m),$$

where the right side glues over $l^\top q$, holds whenever l is a bijection. It fails for some constant kernels l and for some non-injective functions l , so squares are not closed under horizontal composition. The difference of the two sides is always a circulation.

Proof. A bijection relabels the middle variable, which does not change the glued law. For a counterexample with a function, let $p = q = r$ be uniform on $\{0, 1\}$, let $\alpha = \beta$ be the diagonal coupling, let k, m be identities and let l collapse $\{0, 1\}$ to a point. The left side is the diagonal coupling, since gluing two copies of the identity gives the identity. On the right, $k^\top\alpha l$ and $l^\top\beta m$ are product couplings with a one-point middle space, so their gluing is the product coupling $\frac{1}{4}\mathbf{1}\mathbf{1}^\top$. For a genuinely stochastic kernel, take instead l with both rows uniform on $\{0, 1\}$: then αl and $l^\top\beta$ are again product couplings, so the right side is still $\frac{1}{4}\mathbf{1}\mathbf{1}^\top$. Both sides have the margins $k^\top p$ and $m^\top r$, so their difference is a circulation. $\square$

So the structure is a double category, strict and thin, when the vertical arrows are restricted to bijections. With all kernels, horizontal composition of squares is not defined in general, and since squares are a property there are no comparison cells that could make it lax. The defect is a circulation of the same kind as the whiskering defect, so a double-category formulation does not remove the obstruction of [2, Thm. 7.6(v)].

6 The ring of contingency tables

The gluing algebra forgets the geometry of couplings. A different algebra keeps it. Let f, g be rational margins on $\{1, \dots, n\}$ and $\{1, \dots, m\}$ with common denominator N , so that Nf and Ng are integer vectors with the same total. The polytope $P = N \cdot \text{Cpl}(f, g)$ consists of the nonnegative real $n \times m$ tables with row sums Nf and column sums Ng . Its lattice is the set of integer tables, and the lattice points of kP are the integer tables with margins kNf and kNg .

Theorem 6.1 (The ring of contingency tables). *(i) P is a lattice polytope of dimension $(n - 1)(m - 1)$ when f, g have full support, and it has the integer decomposition property: every integer table in kP is a sum of k integer tables in P .*

(ii) The Ehrhart ring $S_P = \bigoplus_{k \geq 0} \mathbb{R}[kP \cap \mathbb{Z}^{n \times m}]$, graded by k with multiplication given by adding tables, is generated in degree one. Its Hilbert function $k \mapsto \dim S_{P,k}$ is the Ehrhart polynomial of P , of degree $(n - 1)(m - 1)$, with constant term 1 and leading coefficient the volume of P normalized to the lattice of integer circulations.

(iii) $\text{Proj } S_P$ is the projective toric variety of P , embedded projectively normally. For nondegenerate margins it is the smooth toric variety $X_{f,g}$ of [3, Thm. 6.4], and S_P is the ring of sections of the powers of the ample line bundle of P , whose class in $\text{Pic}(X_{f,g}) \otimes \mathbb{Q}$ is N times the class of the product coupling.

(iv) The degree-one basis $P \cap \mathbb{Z}^{n \times m}$ is the set of integer tables with margins Nf and Ng , the sample space of Fisher's exact test, on which the conditional law of a table given its margins under independence is the multivariate hypergeometric law, proportional to $1/\prod_{ij} t_{ij}!$. The basic moves $\delta_{ij} + \delta_{i'j'} - \delta_{ij'} - \delta_{i'j}$ connect this set, and each is an integer circulation.

Proof. (i) The constraint matrix of a transportation polytope is the incidence matrix of a complete bipartite graph, which is totally unimodular, so the vertices of P are integral [12], and polyhedra defined by totally unimodular systems with integral right-hand sides have the integer decomposition property [7]. The dimension is that of the circulation space. (ii) Generation in degree one is (i). The Hilbert function counts lattice points of kP , and the properties of the Ehrhart polynomial are standard [8]. (iii) A lattice polytope with the integer decomposition property is normal, hence very ample, and Proj of its Ehrhart ring is the toric variety of its normal fan, embedded projectively normally, and the sections of the k -th power of its line bundle have the lattice points of kP as a basis [9, Ch. 2 and Prop. 4.3.3]. The normal fan of P is that of $\text{Cpl}(f, g)$, which for nondegenerate margins is the smooth fan of $X_{f, g}$. The support numbers of P are N times the entries of the product coupling, which give the class of $\text{Cpl}(f, g)$ [3, Prop. 8.1]; an integral representative comes from any lattice point of P . (iv) The hypergeometric law is classical [1]. For two-way tables the basic moves form a Markov basis [10]; each has zero row and column sums. $\square$

Example 6.2. For $f = (\frac{1}{2}, \frac{1}{2})$ and $g = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, take $N = 6$. The tables with margins $(3, 3)$ and $(2, 2, 2)$ are 7 lattice points of a hexagon. The numbers of tables with margins $k(3, 3)$ and $k(2, 2, 2)$ are 7, 19, 37, 61, 91 for $k = 1, \dots, 5$, equal to $3k^2 + 3k + 1$. The leading coefficient 3 is the lattice area of the hexagon, which is $N^2 = 36$ times the area $\frac{1}{12}$ of $\text{Cpl}(f, g)$ in [3, §8]. For margins $(5, 7)$ and $(3, 4, 5)$ the counts give $10k^2 + 6k + 1$, and the area of that polygon is 10.

Example 6.3 (Fisher's exact test on synthetic data). The first 40 items of the synthetic moderation data of [3, §12] give the label-by-action table

$$T = \begin{pmatrix} 10 & 8 & 1 \\ 2 & 5 & 14 \end{pmatrix}, \quad \text{margins } (19, 21) \text{ and } (12, 13, 15).$$

Its fibre, the degree-one basis of the ring of $\frac{1}{40}$ -these margins, has 151 tables. The hypergeometric probabilities sum to 1, and the exact p -value, the total probability of tables no more likely than T , is 9.8×10^{-5} . The χ^2 approximation with 2 degrees of freedom gives 1.8×10^{-4} .

The three appearances of the circulations now line up. They are the ideal I_f of the gluing algebra (section 2), the tangent space of the coupling polytope, whose dimension $(n-1)(m-1)$ is both the degree of the Ehrhart polynomial and the number of degrees of freedom of the test of independence [3, Prop. 12.1], and the lattice of Markov moves that connects the sample space of the exact test. The first appearance is homotopically trivial (theorem 4.2). The other two carry the geometry, because they keep positivity: a table must stay nonnegative under a move, and the polytope is cut out by nonnegativity.

7 Open problems

1. *The free cone at chain level.* In the free cone of I_f , products of homotopies survive, and the functor of theorem 4.1 acquires arity-four terms. Are the higher components given by the higher Boolean cumulants of whiskering? By theorem 4.2 any answer is invisible up to homotopy.
2. *A positive homotopy theory.* Corollary 4.3 locates contextuality in positivity. Find a homotopy theory of cones of measures, for example through simplicial distributions, in which unfillable horns of the coupling nerve [3, §4] are nontrivial classes.
3. *Ehrhart data of coupling polytopes.* The h^* -vectors of $N \cdot \text{Cpl}(f, g)$ as functions of the chamber of (f, g) , and the asymptotic number of tables near a wall, where the polytope stops being simple [3, §6].

4. *Sections and potentials.* On $X_{f,g}$ the degree-one sections are indexed by tables. Relate the exact test to the disc potential of [3, §7], for example through the leximin coupling, which lies over the Floer-nontrivial fibres at the sampled points of [3, Obs. 10.10].

8 Computational checks

`check_coupling_algebra.py` runs ten checks, all passing:

- A1–A2: [theorem 2.2](#) for $n = 2, \dots, 5$, and [proposition 2.3](#) on objects of sizes 2, 3, 4;
- A3: [proposition 3.1](#);
- A4: [theorem 4.1](#) for convex combinations of the two actions;
- A5: [theorem 4.2](#);
- A6: [proposition 5.1](#);
- A7–A9: [theorem 6.1](#)(i), (ii) and (iv) on two 2×3 examples and a 3×3 example;
- A10: [example 6.3](#), in floating point.

Everything except A10 uses exact rational arithmetic.

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